



UNIVERSITÀ
DEGLI STUDI
DI TRIESTE

PhotoElectrocatalytic smart Systems for CHEmicals and FUEls production

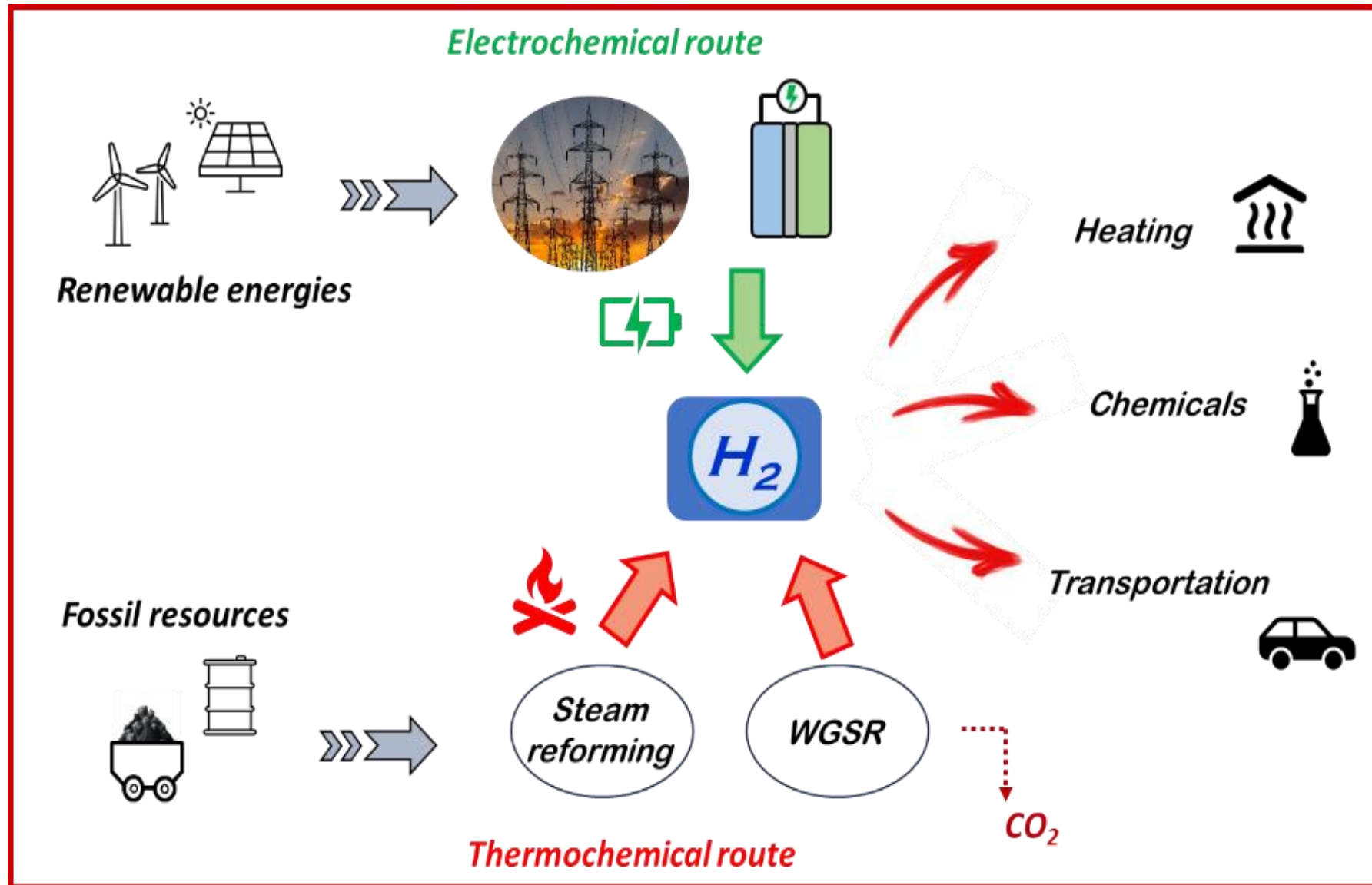
Michele Melchionna
melchionnam@units.it



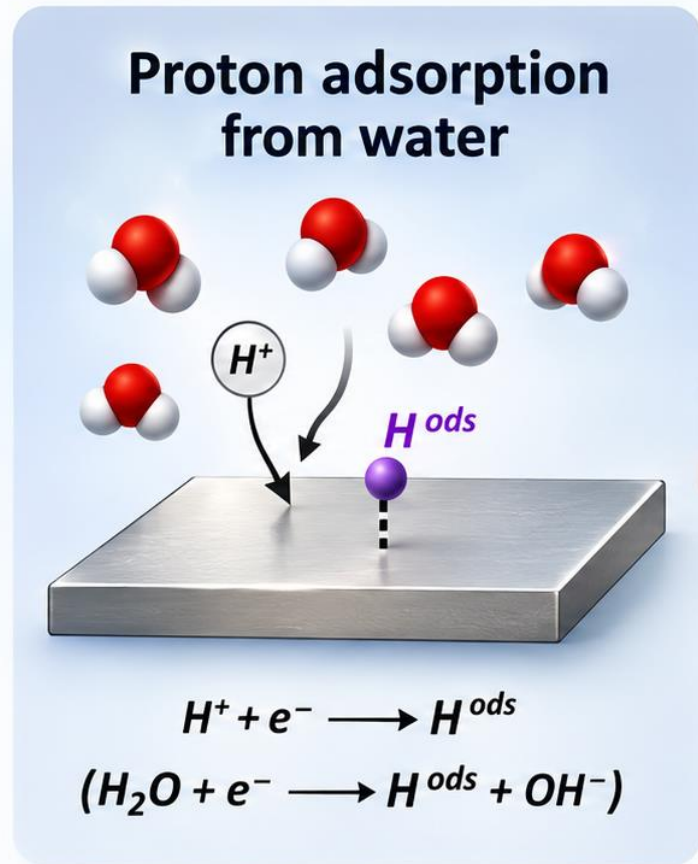
PESCHE-FU

<https://pesche-fu.github.io/index.html>

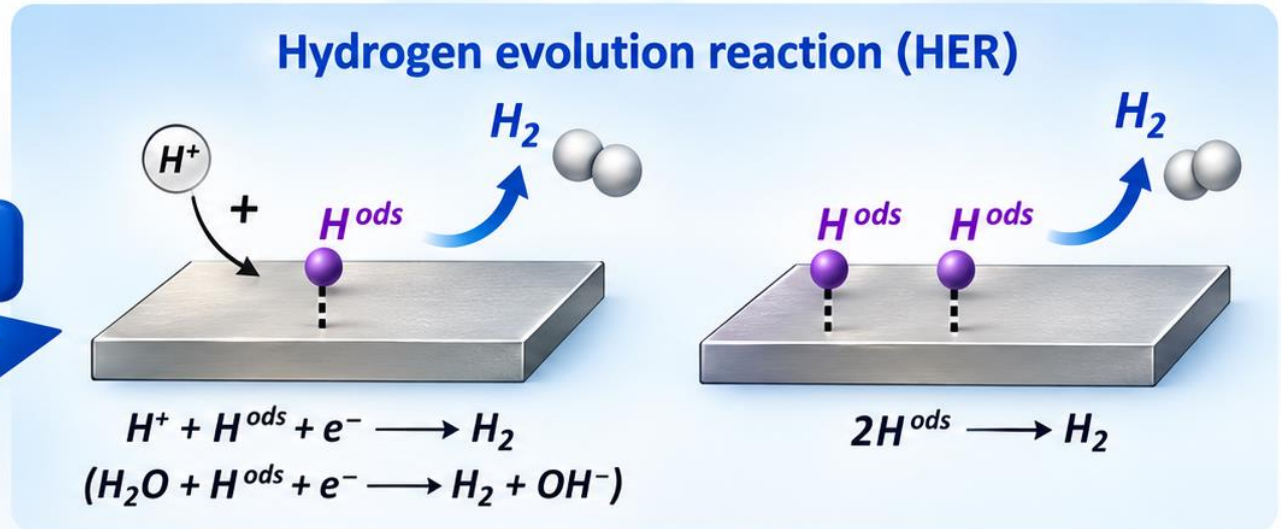
The central role of H₂ in industry



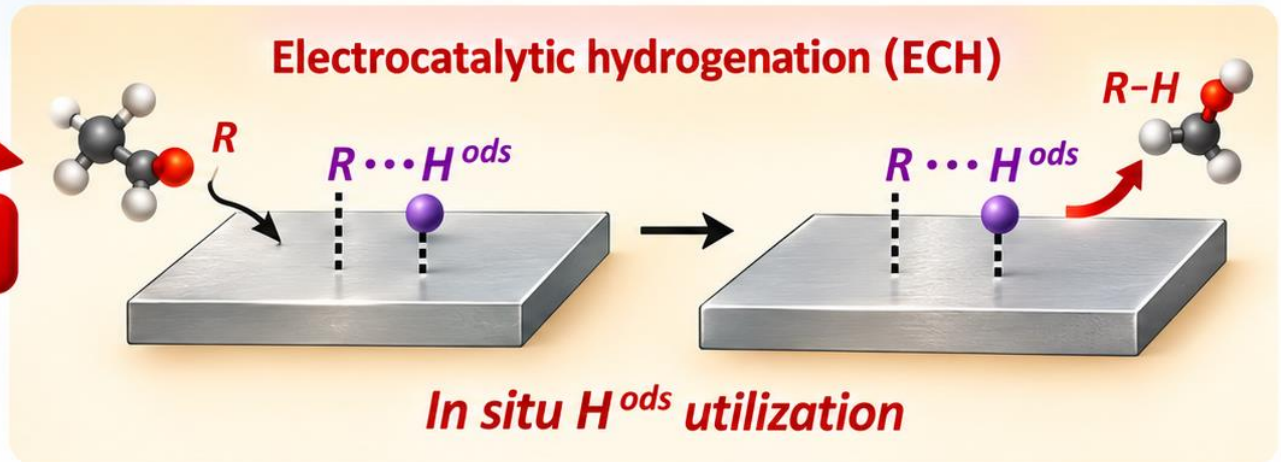
Electrocatalytic H₂ production and in situ utilization



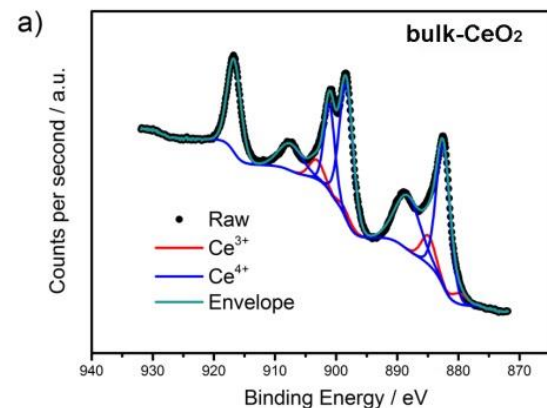
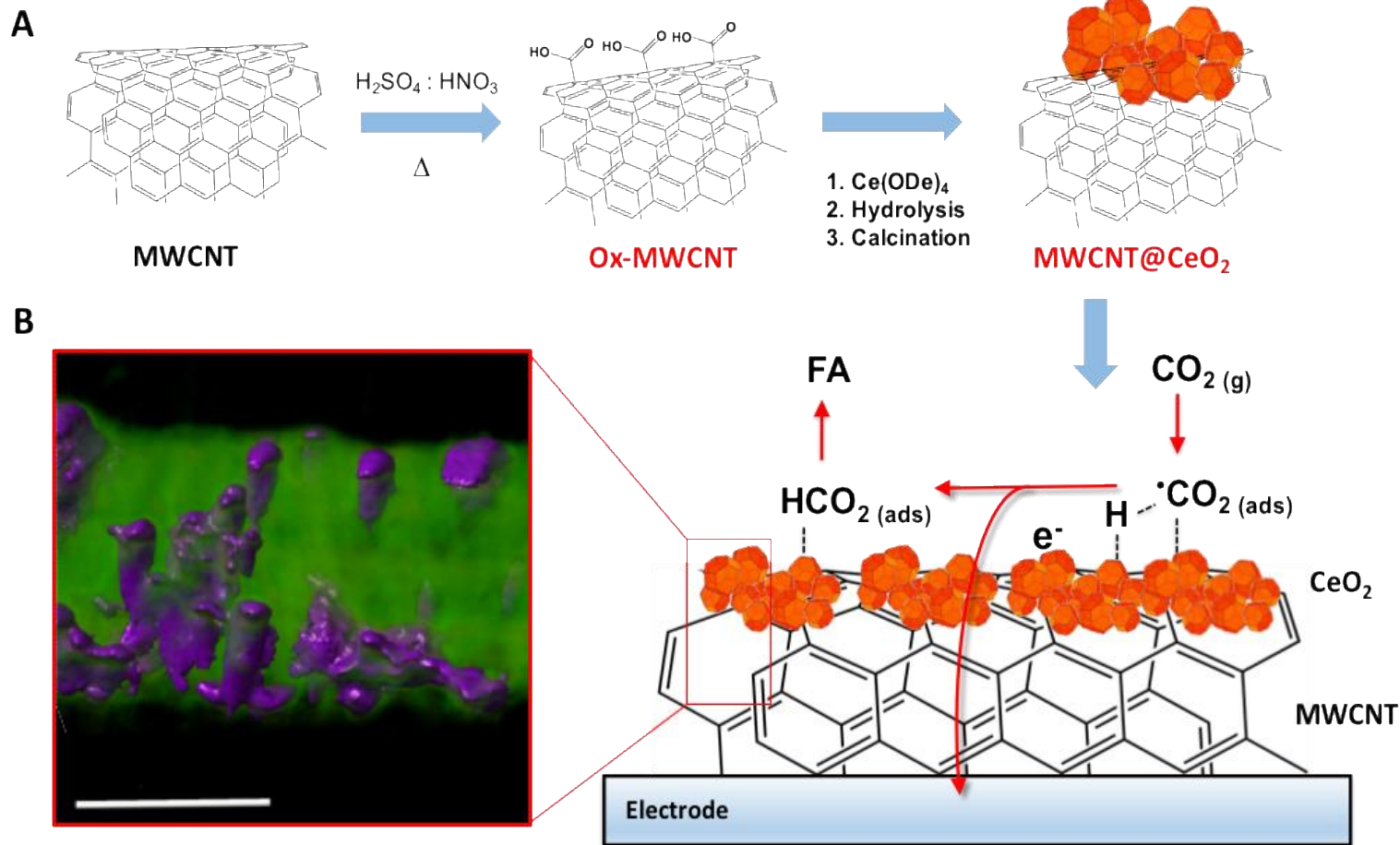
HER



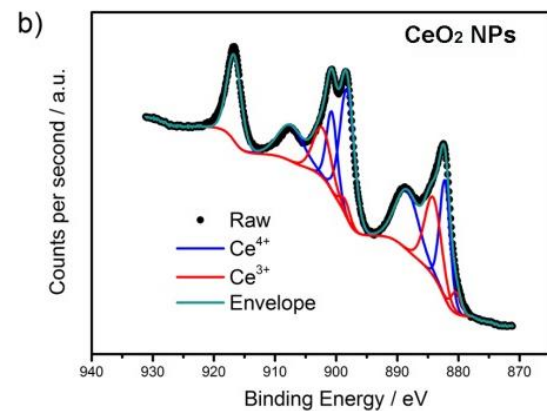
ECH



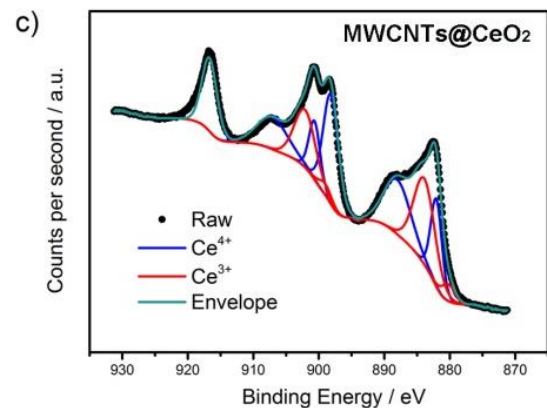
CO₂ electrohydrogenation by CeO₂



Ce³⁺ = 11%

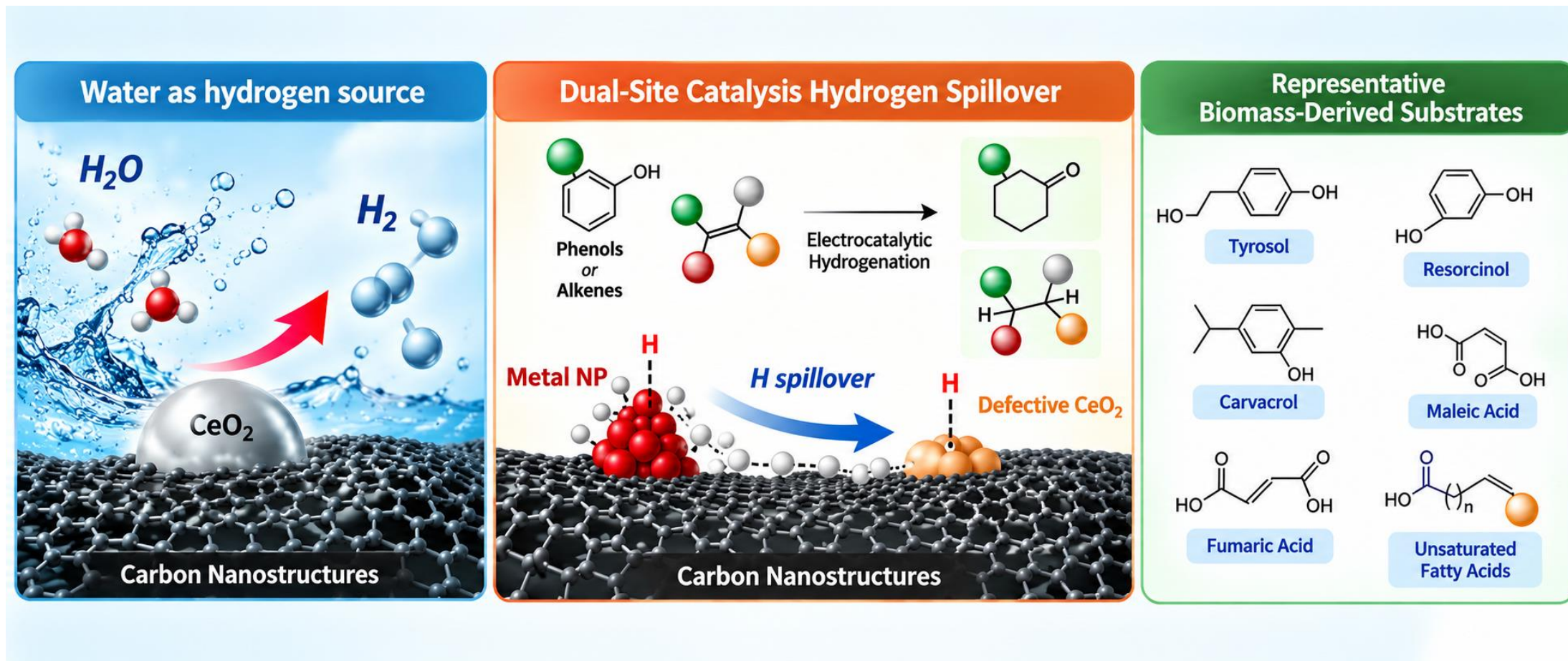


Ce³⁺ = 25%



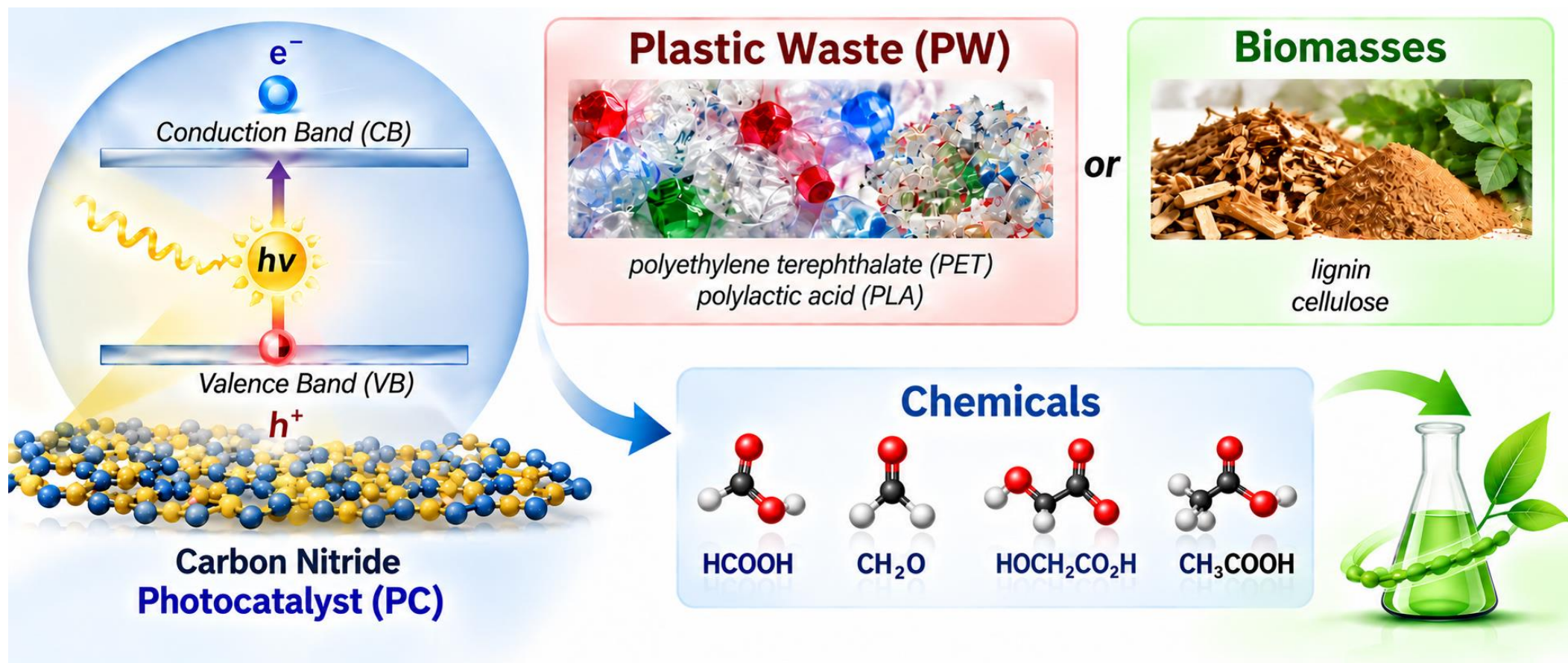
Ce³⁺ = 33%

Organic compounds electrohydrogenation by CeO₂



Li, W.; Gan, J.; Liu, Y.; Zou, Y.; Zhang, S.; Qu, Y., *Angew. Chem. Int. Ed.* **2023**, 62 (37), e202305661

Photocatalysis for organic conversions

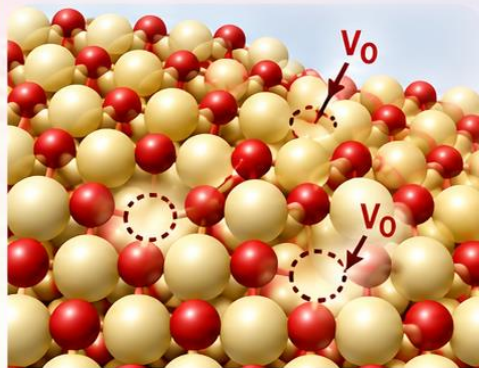


Photoelectrochemical fuel and chemical production

PhotoElectrocatalytic smart Systems for CHEmicals and FUEls production: PESCHE-

FU

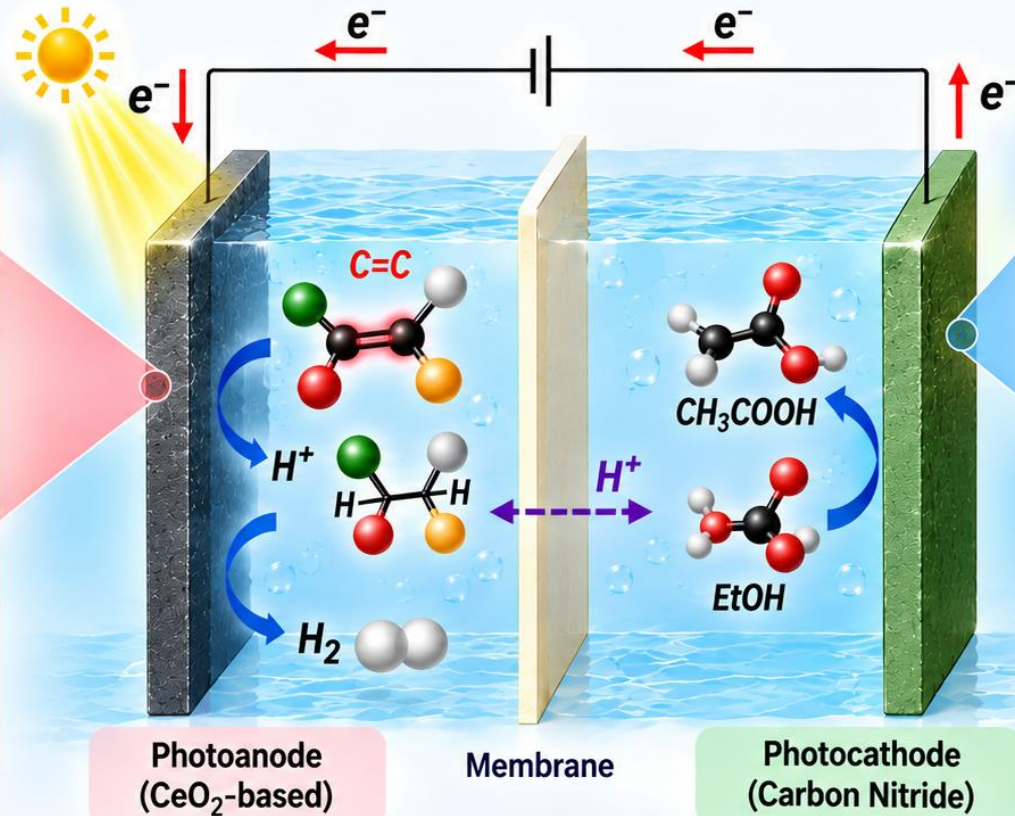
Nanostructured
 CeO_2 -Based Materials



● Ce ● O ○ Oxygen vacancy (V_o)



CeO_2 nanostructures

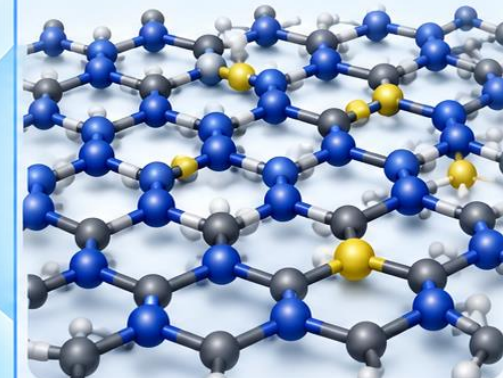


Photoanode
(CeO_2 -based)

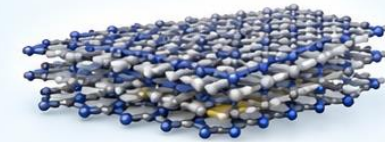
Membrane

Photocathode
(Carbon Nitride)

Carbon Nitrides



● C ● N ● H ● Defect site



$g\text{-C}_3\text{N}_4$ nanosheets

The Team



PRINCIPAL INVESTIGATOR

Prof. Paolo Fornasiero



Prof. Michele Melchionna

Associate Professor



Prof. Federico Franco

Associate Professor



Prof. Giacomo Filippini

Associate Professor



Prof. Tiziano Montini

Associate Professor



Dr. Luca Schio

Assistant Professor (RTT)